

FCC3 222: Fluid Mechanics 1A

Academic Year 2024/2025 - Semester Two

Dept of Civil & Construction Engineering,
University of Nairobi

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Course Content

- Introduction to fluid mechanics; properties of fluids, dimensions and units.
- Hydrostatic pressure, pressure gauges and manometers.
- Forces and centres of pressure of plans and non-plana surfaces.
- Floating bodies, metacentre.
- Free surface correction and suspended loads.
- Introduction to Bernoulli's theorem.

Mode of Delivery

- Largely physical classes however may be blended with online lectures
- 4 Hours of lectures and tutorials per week

The course shall be examined as follows:

1. 2 Hour written Examination at the end of the semester (70%)
2. Continuous Assessment Test, and Assignments (30%)
3. Pass mark 40%

NB: Mobile phones put on silent mode but you may be free to step out and answer **very necessary** phone calls in physical classes and mics muted unless when asking or answering questions in online classes.

References/Reading Materials

- R.S Khurmi, *Hydraulics, Fluid Mechanics and Hydraulic Machines*, S.Chand, New Delhi, 2008
- Robert W. Fox, Philip J. Pritchard, Alan T. McDonald, *Introduction to Fluid Mechanics* 7th Edition, John Wiley, India, 2010
- Er.R.K Bansal, *Fluid Mechanics and Hydraulic Machines*, Revised Ninth Edition, Laxmi Publications, New Delhi, 2013
- Dr.R.K Rajput, *Fluid Mechanics and Hydraulic Machines*, 5th Edition, S.Chand, New Delhi, 2014
- Yunus A. Cengel, John M. Cimbala, *Fluid Mechanics: Fundamentals and Applications* 3rd Edition, McGraw-Hill, 2014
- Doulas,J.F (Ed.) (2011). *Fluid Mechanics* (6th ed). Prentice Hall.ISBN-13: 978-0273717720.
- Batchelor,G.K. (2010). *An Introduction to fluid dynamics* (1.Cambridge mathematical ed., 14. print) Cambridge Univ. Press. ISBN-13: 978-0521663960
- Internet sources
- Journals
- Lecture notes
- Any other relevant materials to the course

LESSON 1: INTRODUCTION

Fluid: A fluid is a substance which is capable of flowing or It is a substance which deforms continuously when subjected to external shearing force.

Fluids have the following characteristics:

- It has no definite shape of its own, but conforms to the shape of the containing vessel
- Even a small amount of shear force exerted on a liquid/fluid will cause it to undergo a deformation which continues as long as the force continues to be applied.

Fluid mechanics: Is a branch of engineering science that deals with the behavior of fluids (liquid and gases) at rest as well as in motion.

The study of the fluid behavior may be categorized as static, kinematics and dynamic.

- Statics is the study of the fluids in a state of rest.

Introduction cont.

- The study of the fluid in motion where pressure forces are not considered is referred to as fluid kinematics, while fluid dynamics on the other hand studies the motion of the fluid under application of pressure forces.

Classification of fluids

Fluids may be classified as follows:

- ❖ Liquid, gas, vapour (Forms of fluids)
- ❖ Ideal fluids or Real fluids,
- ❖ Newtonian fluid or Non-Newtonian fluid,
- ❖ Ideal plastic fluid

- ⑩ **A liquid** is a fluid which possesses a definite volume (which may vary very slightly with temperature and pressure)
- ⑩ **Gas:** It possesses no definite volume and is compressible.
- ⑩ **Vapour:** It is a gas whose temperature and pressure are such that it is very near the liquid state (e.g., steam).

Introduction cont.

- **Real fluids:** A real practical fluid is one which has viscosity, surface tension and compressibility in addition to the density. Such fluids are actually available in nature.
- **Ideal fluids:** Are fluids with no viscosity and surface tension and is incompressible. In true sense no such fluid exists in nature. However, fluids with low viscosities such as water and air can be treated as ideal fluids under certain conditions. The assumption of ideal fluids helps in simplifying the mathematical analysis.
- **Newtonian fluid:** A real fluid in which the shear stress is directly proportional to the rate of shear strain (or velocity gradient). Examples; water, kerosine, air
- **Non-Newtonian Fluid:** A real fluid in which shear stress is not proportional to the rate of shear strain (or velocity gradient). Examples; solutions, mud flows, blood etc.
- **Ideal Plastic Fluid:** A fluid, in which shear stress is more than the yield value and shear stress is proportional to the rate of shear strain (velocity gradient). Examples; sewage, sludge, drilling muds etc

Illustration of types of fluids

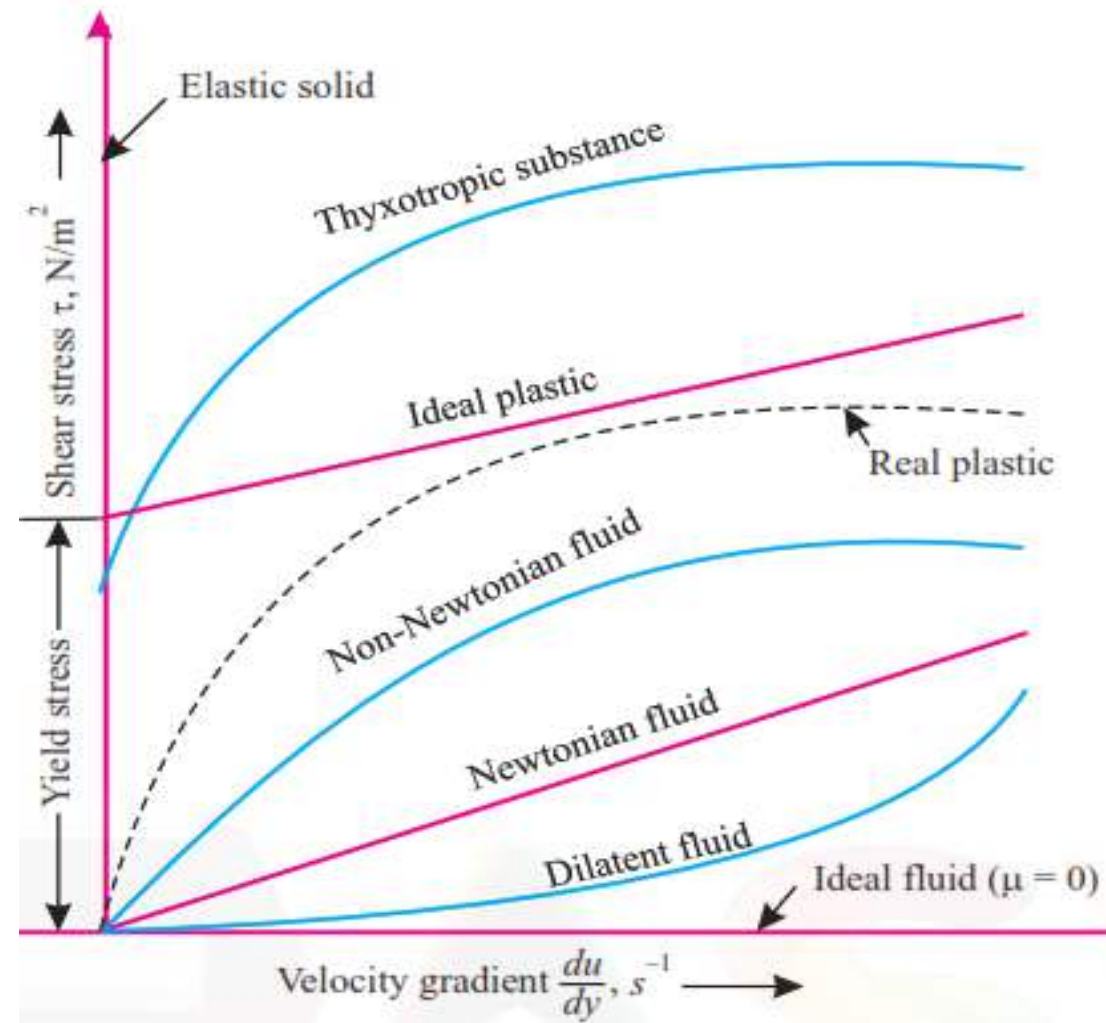


Fig. 1: Types of fluids under variation of shear stress with velocity gradient

Worked Examples

Problem 1.4 A plate 0.025 mm distant from a fixed plate, moves at 60 cm/s and requires a force of 2 N per unit area i.e., 2 N/m² to maintain this speed. Determine the fluid viscosity between the plates.

Solution. Given :

$$\begin{aligned} \text{Distance between plates, } dy &= .025 \text{ mm} \\ &= .025 \times 10^{-3} \text{ m} \end{aligned}$$

$$\text{Velocity of upper plate, } u = 60 \text{ cm/s} = 0.6 \text{ m/s}$$

$$\text{Force on upper plate, } F = 2.0 \frac{\text{N}}{\text{m}^2}.$$

This is the value of shear stress i.e., τ

Let the fluid viscosity between the plates is μ .

$$\text{Using the equation (1.2), we have } \tau = \mu \frac{du}{dy}.$$

$$\text{where } du = \text{Change of velocity} = u - 0 = u = 0.60 \text{ m/s}$$

$$dy = \text{Change of distance} = .025 \times 10^{-3} \text{ m}$$

$$\tau = \text{Force per unit area} = 2.0 \frac{\text{N}}{\text{m}^2}$$

$$\begin{aligned} \therefore 2.0 &= \mu \frac{0.60}{.025 \times 10^{-3}} \quad \therefore \mu = \frac{2.0 \times .025 \times 10^{-3}}{0.60} = 8.33 \times 10^{-5} \frac{\text{Ns}}{\text{m}^2} \\ &= 8.33 \times 10^{-5} \times 10 \text{ poise} = \mathbf{8.33 \times 10^{-4} \text{ poise. Ans.}} \end{aligned}$$

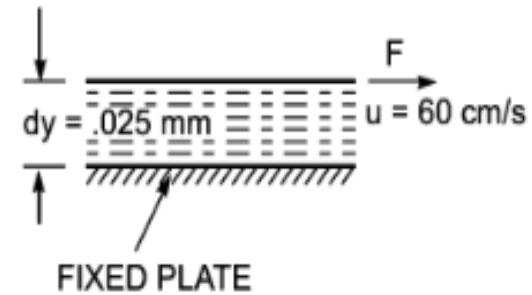


Fig. 1.3

Properties of Fluids

- The following are the important properties of fluids:
 - i) Density
 - ii) Specific gravity
 - iii) Viscosity
 - iv) Vapour pressure
 - v) Cohesion
 - vi) Adhesion
 - vii) Surface tension
 - viii) Capillarity
 - ix) Compressibility

1. Density

Density/Mass Density – Is defined as the ratio of the mass of a fluid to its volume. Thus, it is the mass per unit volume of the fluid. The density of a fluid may be considered constant while that of gases change with variation in pressure and temperature.

Mathematically, density is expressed as:

$$\rho = \frac{\text{Mass of fluid}}{\text{Volume of fluid}} .$$

The value of density of water is 1 gm/cm^3 or 1000 kg/m^3 .

2. Specific Weight

This is defined as the ratio between the weight of a fluid to its volume. Thus, weight per unit volume of a fluid is referred to as specific weight or weight density denoted by the symbol (w)

Mathematically expressed as:

$$\begin{aligned}w &= \frac{\text{Weight of fluid}}{\text{Volume of fluid}} = \frac{(\text{Mass of fluid}) \times \text{Acceleration due to gravity}}{\text{Volume of fluid}} \\&= \frac{\text{Mass of fluid} \times g}{\text{Volume of fluid}} \\&= \rho \times g\end{aligned}\quad \left\{ \because \frac{\text{Mass of fluid}}{\text{Volume of fluid}} = \rho \right\}$$

Thus: $w = \rho g$

3. Specific Gravity

Specific gravity is the ratio of the specific weight of the liquid to the specific weight of a standard fluid. It is dimensionless and has no units. It is represented by S .

For liquids, the standard fluid is pure water at 4°C. Specific weight of liquid

$$\therefore \text{Specific gravity} = \frac{\text{Specific weight of liquid}}{\text{Specific weight of pure water}}$$

Example: Calculate the specific weight, specific mass, specific volume and specific gravity of a liquid having a volume of 6 m³ and weight of 44 kN.

Solution: Volume of the liquid = 6 m³
Weight of the liquid = 44 kN

Specific weight, w :

$$w = \frac{\text{Weight of liquid}}{\text{Volume of liquid}} = \frac{44}{6} = 7.333 \text{ kN/m}^3 \text{ (Ans.)}$$

Specific mass or mass density, ρ :

$$\rho = \frac{w}{g} = \frac{7.333 \times 1000}{9.81} = 747.5 \text{ kg/m}^3 \text{ (Ans.)}$$

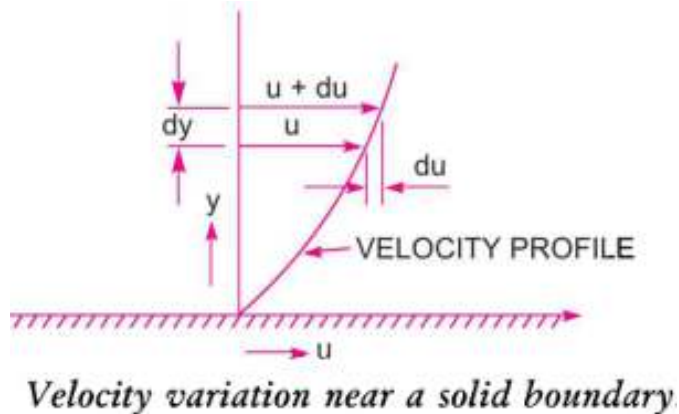
$$\text{Specific volume, } v = \frac{1}{\rho} = \frac{1}{747.5} = 0.00134 \text{ m}^3/\text{kg} \text{ (Ans.)}$$

Specific gravity, S :

$$S = \frac{w_{\text{liquid}}}{w_{\text{water}}} = \frac{7.333}{9.81} = 0.747 \text{ (Ans.)}$$

4. Viscosity

Viscosity is defined as the property of a fluid which offers resistance to the movement of one layer of fluid over another adjacent layer of the fluid. Viscosity of fluids is caused by cohesion and interaction between particles. When two layers of a fluid, a distance ' dy ' apart move one over the other at different velocities, say u and $v+du$ as shown in Fig. below, the viscosity together with relative velocities causes a shear stress acting between the fluid layers.



The top layer causes a shear stress on the adjacent lower layer while the lower layer causes a shear stress on the adjacent top layer. This shear stress is proportional to the rate of change of velocity with respect to y . It is denoted by the symbol τ (Tau).

Mathematically,

$$\tau \propto \frac{du}{dy}$$

$$\tau = \mu \frac{du}{dy}$$

Viscosity

Where; μ (mu) is the constant of proportionality known as the co-efficient of dynamic viscosity whereas $\frac{du}{dy}$ represents the rate of shear strain or rate of shear deformation or velocity gradient.

Units of Viscosity:

In *S.I. Units*: N.s/m^2

In *M.K.S. Units*: $\text{kg}_f\text{sec/m}^2$

$$\left[\because \mu = \frac{\text{force/area}}{(\text{length/time}) \times \frac{1}{\text{length}}} = \frac{\text{force/length}^2}{\frac{1}{\text{length}}} = \frac{\text{force} \times \text{time}}{(\text{length})^2} \right]$$

The unit of viscosity in C.G.S. is also called *poise* = $\frac{\text{dyne} - \text{sec}}{\text{cm}^2}$. One poise = $\frac{1}{10} \text{N.s/m}^2$

Kinematic Viscosity – is the ratio between the dynamic viscosity and density of a fluid, denoted by ν (nu).

Mathematically, $\nu = \frac{\text{Viscosity}}{\text{Density}} = \frac{\mu}{\rho}$

Viscosity

Worked Examples:

A plate 0.025 mm distant from a fixed plate, moves at 60 cm/s and requires a force 2 N/m² to maintain this speed. Determine the fluid viscosity between the plates.

Solution. Given :

$$\begin{aligned}\text{Distance between plates, } dy &= .025 \text{ mm} \\ &= .025 \times 10^{-3} \text{ m}\end{aligned}$$

$$\text{Velocity of upper plate, } u = 60 \text{ cm/s} = 0.6 \text{ m/s}$$

$$\text{Force on upper plate, } F = 2.0 \frac{\text{N}}{\text{m}^2}.$$

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Let the fluid viscosity between the plates is μ .

Using the equation (1.2), we have $\tau = \mu \frac{du}{dy}$.

$$\text{where } du = \text{Change of velocity} = u - 0 = u = 0.60 \text{ m/s}$$

$$dy = \text{Change of distance} = .025 \times 10^{-3} \text{ m}$$

$$\tau = \text{Force per unit area} = 2.0 \frac{\text{N}}{\text{m}^2}$$

$$\begin{aligned}\therefore 2.0 &= \mu \frac{0.60}{.025 \times 10^{-3}} \quad \therefore \mu = \frac{2.0 \times .025 \times 10^{-3}}{0.60} = 8.33 \times 10^{-5} \frac{\text{Ns}}{\text{m}^2} \\ &= 8.33 \times 10^{-5} \times 10 \text{ poise} = \mathbf{8.33 \times 10^{-4} \text{ poise. Ans.}}\end{aligned}$$

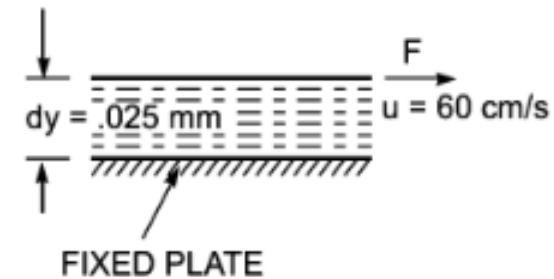


Fig. 1.3

Worked Examples:

The space between two square flat parallel plates is filled with oil. Each side of the plate is 720 mm. The thickness of the oil film is 15 mm. The upper plate, which moves at 3 m/s requires a force of 120 N to maintain the speed. Determine

- The dynamic viscosity of the oil
- The kinematic viscosity of oil if the specific gravity of oil is 0.95.

Solution. Each side of a square plate = 720 mm = 0.72 m

The thickness of the oil, $dy = 15 \text{ mm} = 0.015 \text{ m}$

Velocity of the upper plate = 3 m/s

$\therefore$ Change of velocity between plates, $du = 3 - 0 = 3 \text{ m/s}$

Force required on upper plate, $F = 120 \text{ N}$

$$\therefore \text{Shear stress, } \tau = \frac{\text{force}}{\text{area}} = \frac{120}{0.72 \times 0.72} = 231.5 \text{ N/m}^2$$

(i) Dynamic viscosity, μ :

We know that,

$$\tau = \mu \cdot \frac{du}{dy}$$
$$231.5 = \mu \cdot \frac{3}{0.015}$$

$$\therefore \mu = \frac{231.5 \times 0.015}{3} = 1.16 \text{ N.s/m}^2 \text{ (Ans.)}$$

(ii) Kinematic viscosity, ν :

Weight density of oil, $w = 0.95 \times 9.81 \text{ kN/m}^3 = 9.32 \text{ kN/m}^3 = \text{or } 9320 \text{ N/m}^3$

$$\text{Mass density of oil, } \rho = \frac{w}{g} = \frac{9320}{9.81} = 950$$

$$\text{Using the relation: } \nu = \frac{\mu}{\rho} = \frac{1.16}{950} = 0.00122 \text{ m}^2/\text{s}$$

$$\text{Hence } \nu = 0.00122 \text{ m}^2/\text{s} \text{ (Ans.)}$$

Worked Examples:

Calculate the dynamic viscosity of an oil, which is used for lubrication between a square plate of size 0.8 m x 0.8 m and an inclined plane with an angle of inclination 30° as shown in Fig. below. The weight of the square is 300 N and it slides down the inclined plane with a uniform velocity of 0.3 m/s. The thickness of oil film is 1.5 mm.

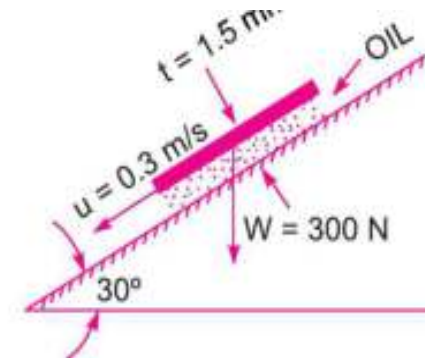
Solution. Given :

Area of plate, $A = 0.8 \times 0.8 = 0.64 \text{ m}^2$

Angle of plane, $\theta = 30^\circ$

Weight of plate, $W = 300 \text{ N}$

Velocity of plate, $u = 0.3 \text{ m/s}$



Thickness of oil film, $t = dy = 1.5 \text{ mm} = 1.5 \times 10^{-3} \text{ m}$

Let the viscosity of fluid between plate and inclined plane is μ .

Component of weight W , along the plane $= W \sin 30^\circ = 300 \sin 30^\circ = 150 \text{ N}$

Thus the shear force, F , on the bottom surface of the plate $= 150 \text{ N}$

and shear stress, $\tau = \frac{F}{\text{Area}} = \frac{150}{0.64} \text{ N/m}^2$

Now using equation (1.2), we have

$$\tau = \mu \frac{du}{dy}$$

where $du = \text{change of velocity} = u - 0 = u = 0.3 \text{ m/s}$

$$dy = t = 1.5 \times 10^{-3} \text{ m}$$

$$\therefore \frac{150}{0.64} = \mu \frac{0.3}{1.5 \times 10^{-3}}$$

$$\therefore \mu = \frac{150 \times 1.5 \times 10^{-3}}{0.64 \times 0.3} = 1.17 \text{ N s/m}^2 = 1.17 \times 10 = \mathbf{11.7 \text{ poise. Ans.}}$$

5. Surface Tension

- **Surface tension** refers to the tensile force acting on the surface of a liquid in contact with a gas or on the surface between two immiscible liquids such that the contact surface behaves like a membrane under tension. Surface tension is caused by the force of cohesion at the free surface.
- **Cohesion** refers to the intermolecular attraction between molecules of the same liquid. It enables a liquid to resist small amount of tensile stress. Cohesion is the tendency of the liquid to remain as one assemblage of particles.
- **Adhesion** – means attraction between the molecules of a liquid and the molecules of a solid boundary surface in contact with the liquid. The property enables a liquid to stick to another body.
- Some phenomenon of surface tension are as follows:

Surface Tension

- i. Rain drops (A falling rain drop becomes spherical due to the cohesion and surface tension)
- ii. Rise of sap in a tree
- iii. Bird drinking water from a pond
- iv. Capillary rise and capillary siphoning
- v. Collection of dust particles on water surface
- vi. Break up of liquid jets

Dimensional Formula for Surface Tension

The dimensional formula is given by: $\left[\frac{E}{L} \right]$ or $\left[\frac{M}{T^2} \right]$

It is usually expressed in N/m. The value of surface tension depends upon the following factors:

- i. Nature of the liquid
- ii. Nature of the surrounding matter (e.g solid, liquid or gas)
- iii. Kinetic energy (and hence the temperature of the liquid molecules)

Surface Tension

Surface Tension on Water Droplet

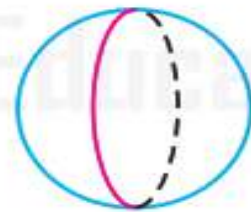
Case I. Water droplet:

Let, p = Pressure inside the droplet above outside pressure (i.e., $\Delta p = p - 0 = p$ above atmospheric pressure)

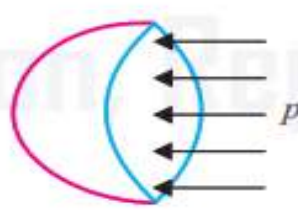
d = Diameter of the droplet and

σ = Surface tension of the liquid.

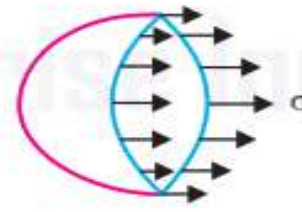
From free body diagram (Fig. 1.19 d), we have:



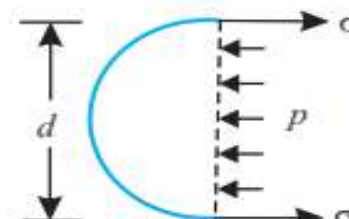
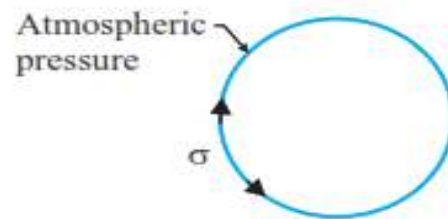
(a) Water droplet



(b) Pressure forces



(c) Surface tension



(d) Free body diagram

(i) Pressure force = $p \times \frac{\pi}{4} d^2$, and

(ii) Surface tension force acting around the circumference = $\sigma \times \pi d$.

Under equilibrium conditions these two forces will be equal and opposite,

$$\text{i.e.,} \quad p \times \frac{\pi}{4} d^2 = \sigma \times \pi d$$

$\therefore$

$$p = \frac{\sigma \times \pi d}{\frac{\pi}{4} d^2} = \frac{4\sigma}{d}$$

This equation shows pressure intensity decreases with increase in size of the droplet

Surface Tension

Surface Tension on Soap (or Hollow) Bubble

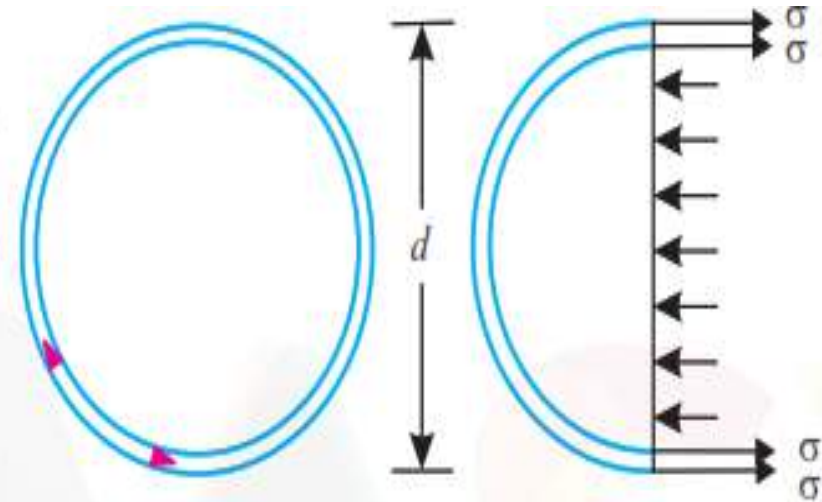
Case II. Soap (or hollow) bubble:

Soap bubbles have two surfaces on which surface tension σ acts.

From the free body diagram (Fig. 1.20), we have

$$p \times \frac{\pi}{4} d^2 = 2 \times (\vec{\sigma} \times \pi d)$$

$$\therefore p = \frac{2\sigma \times \pi d}{\frac{\pi}{4} d^2} = \frac{8\sigma}{d} \quad \dots(1.18)$$



Free body diagram

Fig. 1.20. Pressure inside a soap bubble.

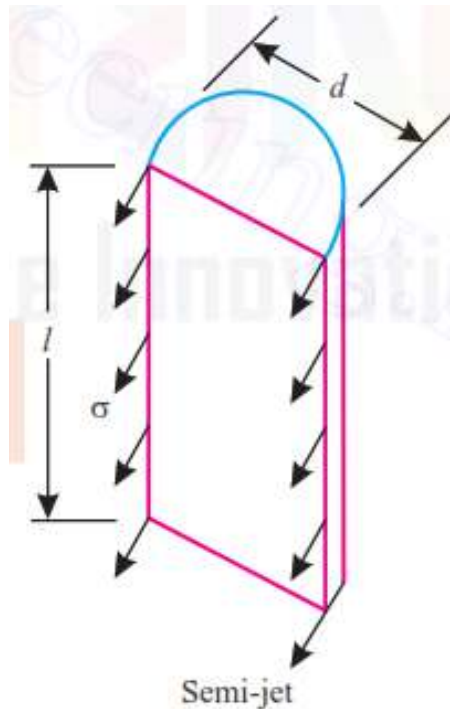
Since the soap solution has a high value of surface tension σ , even with small pressure of blowing a soap bubble will tend to grow larger in diameter (hence formation of large soap bubbles).

Surface Tension

Surface Tension on Soap (or Hollow) Bubble

Case III. A Liquid Jet

Consider a cylindrical liquid jet of diameter d and length l in fig. below of a semi-jet.



$$\text{Pressure force} = p \times l \times d$$

$$\text{Surface tension force} = \sigma \times 2l$$

Equating the two forces, we have:

$$p \times l \times d = \sigma \times 2l$$

$$p = \frac{\sigma \times 2l}{l \times d} = \frac{2\sigma}{d}$$

Worked Examples

1. Find the surface tension in a soap bubble of 40 mm diameter when the inside pressure is 2.5 N/m^2 above the atmospheric pressure.

Solution. Given :

Dia. of bubble, $d = 40 \text{ mm} = 40 \times 10^{-3} \text{ m}$

Pressure in excess of outside, $p = 2.5 \text{ N/m}^2$

For a soap bubble, using equation (1.15), we get

$$p = \frac{8\sigma}{d} \quad \text{or} \quad 2.5 = \frac{8 \times \sigma}{40 \times 10^{-3}}$$

$$\sigma = \frac{2.5 \times 40 \times 10^{-3}}{8} \text{ N/m} = \mathbf{0.0125 \text{ N/m. Ans.}}$$

2. In order to form a stream of bubbles, air is introduced through a nozzle into a tank of water at 20°C . If the process requires 3.0 mm diameter bubbles to be formed, by how much the air pressure at the nozzle must exceed that of the surrounding water?

What would be the absolute pressure inside the bubble if the surrounding water is at 100.3 kN/m^2 ? Take surface tension of water at $20^\circ\text{C} = 0.0735 \text{ N/m}$.

Solution. Diameter of a bubble, $d = 3.0 \text{ mm} = 3 \times 10^{-3} \text{ m}$

Surface tension of water at 20°C , $\sigma = 0.0735 \text{ N/m}$

The excess pressure intensity of air over that of surrounding water, $\Delta p = p$.

We know,

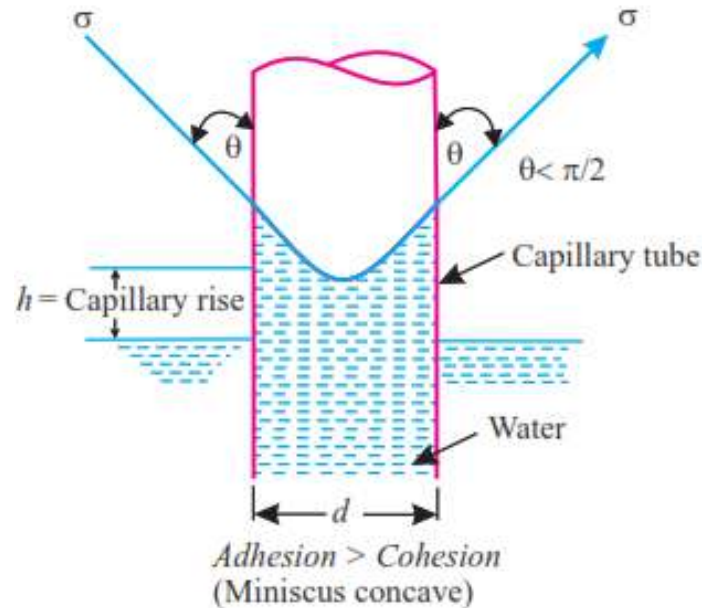
$$p = \frac{4\sigma}{d} = \frac{4 \times 0.0735}{3 \times 10^{-3}} = \mathbf{98 \text{ N/m}^2 \text{ (Ans.)}}$$

Absolute pressure inside the bubble, p_{abs} :

$$\begin{aligned} p_{abs} &= p + p_{atm} \\ &= 98 \times 10^{-3} + 100.3 \\ &= 0.098 + 100.3 = \mathbf{100.398 \text{ kN/m}^2 \text{ (Ans.)}} \end{aligned}$$

6. Capillarity

Capillarity is the phenomenon through which a liquid rises into a thin glass tube above or below its general level. The process results from the combined effect of cohesion and adhesion of liquid particles. Fig. below illustrates the scenario.



The figure illustrates a phenomenon of rising water in the tube of smaller diameters.

Let, d = Diameter of the capillary tube

θ = Angle of contact of the water surface

σ = Surface tension force for unit length, and

w = Weight density (ρg).

From the figure, the upward surface tension force = weight of the water column in the tube (gravity force)

$$\pi d \cdot \sigma \cos \theta = \frac{\pi}{4} d^2 \times h \times w$$

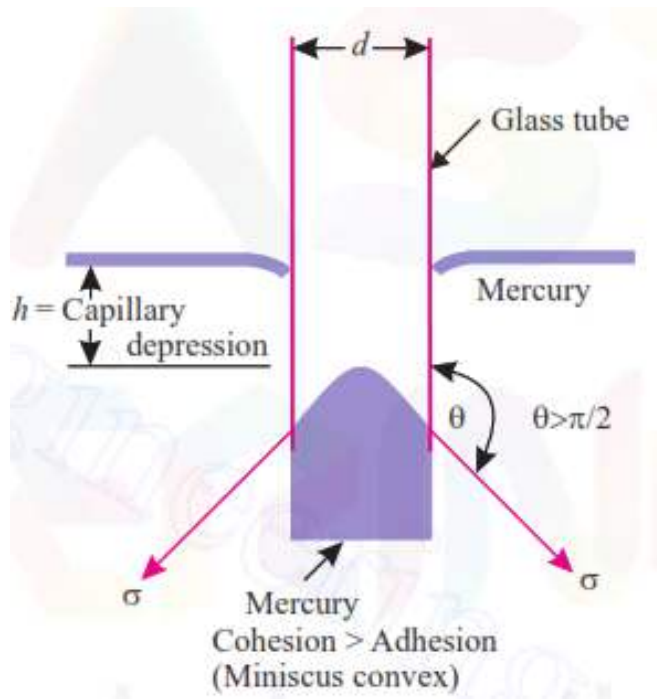
$$\therefore h = \frac{4\sigma \cos \theta}{wd}$$

For water and glass, $\theta \approx 0$, Hence the capillary rise of water in the glass tube,

$$h = \frac{4\sigma}{wd}$$

Capillarity

In case of mercury there is a capillarity depression as shown;



The following points are worth noting:

Smaller the diameter of the capillary tube the greater is the capillary rise or depression

The measurement of liquid level in laboratory capillary (glass) tubes should not be smaller than 8 mm.

Capillary effects are negligible for tubes longer than 12 mm.

For wetting liquid (water), $\theta < \frac{\pi}{2}$. For water, $\theta = 0$ where pure water is in contact with glass.

The effects of surface tension are negligible in many flow problems except those involving.

- ❖ Capillary rise
- ❖ Formation of drops and bubbles
- ❖ The break up of liquid jets and
- ❖ Hydraulic model studies where the model or flow depth is small

Capillarity

Worked Examples

1. Calculate the capillary effect in mm in a glass tube 4 mm diameter, immersed in (i) water and (ii) mercury. The temperature of the liquid is 20°C and the values of surface tension of water and mercury at 20°C in contact with air are 0.0735 N/m and 0.51 N/m respectively. The contact angle of water $\theta=0^\circ$ and t for mercury $\theta=130^\circ$. Take the specific weight of water at 20°C as equal to = 9790 N/m³.

Solution. *Given:* Diameter of glass tube, $d = 4 \text{ mm} = 0.004 \text{ m}$

Surface tension at 20°C, σ :

$$\sigma_{\text{water}} = 0.0735 \text{ N/m}, \quad \sigma_{\text{mercury}} = 0.051 \text{ N/m}$$

Specific weight of water at 20°C = 9790 N/m³

The rise or depression h of a liquid in a capillary tube is given by

$$h = \frac{4\sigma \cos \theta}{wd}$$

where, σ = surface tension, θ = angle of contact, and w = specific weight.

(i) Capillary effect for water:

$$\begin{aligned} h &= \frac{4 \times 0.0735 \times \cos 0^\circ}{9790 \times 0.004} & (\because \theta_{\text{water}} = 0^\circ \dots \text{given}) \\ &= 7.51 \times 10^{-3} \text{ m} = \mathbf{7.51 \text{ mm (rise) (Ans.)}} \end{aligned}$$

(ii) Capillary effect for mercury

$$\begin{aligned} h &= \frac{4 \times 0.051 \times \cos 130^\circ}{(13.6 \times 9790) \times 0.004} & (\because \theta_{\text{mercury}} = 130^\circ \dots \text{given}) \\ &= -2.46 \times 10^{-3} \text{ m} = -2.46 \text{ mm} \\ \text{or,} & \\ \text{i.e.,} & \quad h = \mathbf{2.46 \text{ mm (depression) (Ans.)}} \end{aligned}$$

7. Compressibility and Bulk Modulus

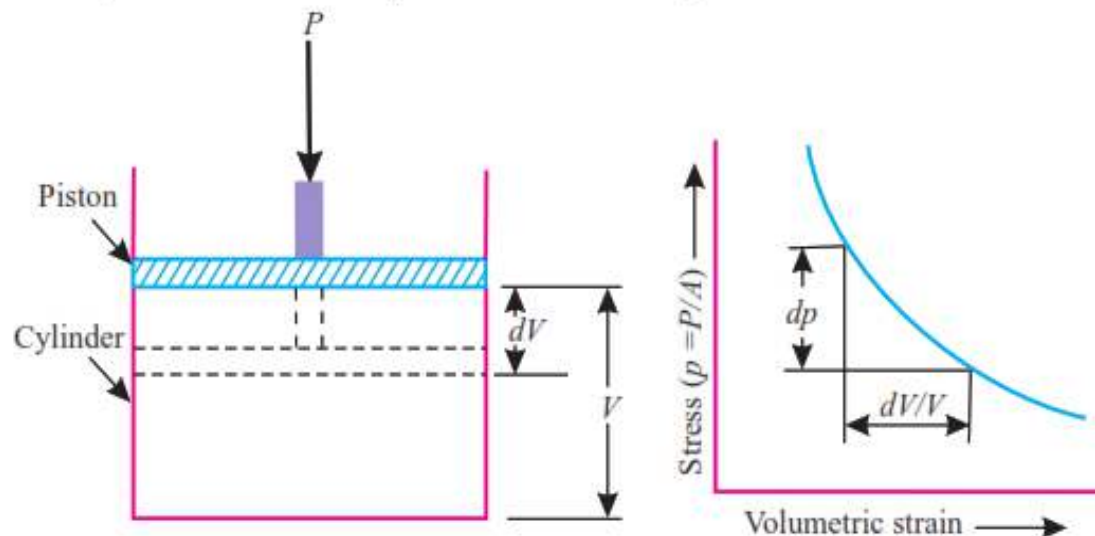
Compressibility is the property of fluids that allows it to undergo a change in volume under the action of external pressure. The property allows the fluid to decrease in volume with increase in pressure.

Fluids like water has very little variation of volume under pressure such that for practical purposes, it is considered incompressible. However, in special scenarios of sudden large changes in pressure such as (water hammer) in pipes, the compressibility cannot be neglected.

Compressibility in fluid mechanics is considered mainly when the velocity of flow is high enough reaching 20% of speed of sound in the medium.

Elasticity of fluids is measured in terms of **bulk modulus of elasticity** (K) which is defined as the ratio of compressive stress to volumetric strain. Compressibility is inversely proportional to the modulus of elasticity.

Consider a cylinder fitted with a piston as shown below.



V = Volume of gas enclosed in the cylinder

P = Pressure of gas when volume is v

$= \frac{P}{A}$ where A is the cross-section area of the cylinder

Let pressure = $p+dp$ and Volume decrease by dv

Compressibility and Bulk Modulus

$$\therefore \text{Volumetric strain} = -\frac{dV}{V}$$

(Negative sign indicates *decrease in volume with increase of pressure*)

$$\therefore \text{Bulk modulus, } K = \frac{dp(\text{increase of pressure})}{-dV/V(\text{volumetric strain})}$$

$$\text{i.e., } K = \frac{dp}{-dV/V}$$
$$\left(\text{Compressibility} = \frac{1}{K} \right)$$

Worked Examples

1. Determine the bulk modulus of elasticity of a liquid, if the pressure of the liquid is increased from 70 N/cm² to 130 N/cm². The volume of the liquid decreases by 0.15%

Solution. Given :

Initial pressure = 70 N/cm²

Final pressure = 130 N/cm²

$\therefore dp = \text{Increase in pressure} = 130 - 70 = 60 \text{ N/cm}^2$

Decrease in volume = 0.15%

$$\therefore -\frac{dV}{V} = +\frac{0.15}{100}$$

Bulk modulus, K is given by equation (1.10) as

$$K = \frac{dp}{-\frac{dV}{V}} = \frac{60 \text{ N/cm}^2}{\frac{.15}{100}} = \frac{60 \times 100}{.15} = 4 \times 10^4 \text{ N/cm}^2. \text{ Ans.}$$

Compressibility and Bulk Modulus

Worked Examples

2. When the pressure of liquid is increased from 3.5 MN/m² to 6 MN/m² its volume is found to decrease by 0.08%. What is the bulk modulus of elasticity of the liquid?

Solution. Initial pressure = 3.5 MN/m²
Final pressure = 6.5 MN/m²
 $\therefore$ Increase in pressure, $dp = 6.5 - 3.5 = 3.0 \text{ MN/m}^2$
Decrease in volume = 0.08 percent $\therefore -\frac{dV}{V} = \frac{0.08}{100}$

Bulk modulus (K) is given by:

$$K = \frac{dp}{-\frac{dV}{V}} = \frac{3 \times 10^6}{\frac{0.08}{100}} = 3.75 \times 10^9 \text{ N/m}^2 \text{ or } 3.75 \text{ GN/m}^2$$

Hence, $K = 3.75 \text{ GN/m}^2$ (Ans.)

8. Vapour Pressure

A change from the liquid state to gaseous state is known as **vaporization**. Vaporization occurs due to continuous escape of the molecules through the free liquid surface.

Vapour pressure is the pressure at which the liquid is converted into vapours

When pressure at any point in flowing liquid becomes equal to or less than the vapour pressure, the vaporization of the liquid begins. The bubbles of these vapours are carried by the flowing liquid into the region of high pressure where they collapse, giving rise to high impact pressure. The pressure developed by collapsing bubbles is so high that the material from the adjoining boundaries gets eroded and cavities are formed on them. This phenomenon is known as **cavitation**.

Cavitation is therefore defined as the phenomenon of formation of bubbles of a flowing liquid in a region where the pressure of the liquid falls below the vapour pressure and sudden collapsing of these vapour bubbles in a region of higher pressure.

Points to note about Vapour Pressure:

1. *If the pressure on the liquid surface is lower than or equal to the saturation vapour pressure, then boiling takes place*
2. *Vapour pressure increases with the rise in temperature*
3. *Mercury has a very low vapour pressure and hence, it is an excellent fluid to be used in a barometer.*

Summary of Fluid Characteristics

Sr. No.	Characteristics	Symbol	Definition	Dimensions	Units
1.	Mass density	ρ	Mass per unit volume, $\frac{m}{V}$	ML^{-3}	kg/m ³
2.	Weight density (or specific weight)	w	Weight per unit volume, $\frac{w}{V}$	FL^{-3}	N/m ³
3.	Specific volume	v	Volume per unit mass $\frac{V}{m} = \frac{1}{\rho}$	$L^3 M^{-1}$	m ³ /kg
4.	Specific gravity	S	$\frac{\text{Specific weight of liquid}}{\text{Specific weight of pure water}}$ $= \frac{w_{\text{liquid}}}{w_{\text{water}}}$		
5.	Dynamic viscosity	μ	Newton's law: $\tau = \mu \cdot \frac{du}{dy}$	FTL^{-2}	N.s/m ² poise, centipoise
6.	Kinematic viscosity	ν	$\nu = \frac{\mu}{\rho}$	$L^2 T^{-1}$	m ² /s stoke, centistoke
7.	Bulk modulus	K	$K = -\frac{\Delta p}{dV/V}$	FL^{-2}	N/m ²
8.	Surface tension	σ	Force per unit length	FL^{-1}	N/m
9.	Vapour pressure	p	$p_v = \frac{F}{A}$	FL^{-2}	N/m ²

Properties of Some Common Fluids at 20°C and Atmospheric Pressure

Fluid	Mass density $\rho(\text{kN/m}^3)$	Specific weight $w(\text{kN/m}^3)$	Dynamic viscosity μ		Kinematic viscosity ν		Modulus of elasticity $E(\text{N/m}^2)$	Surface tension in contact with air, σ (N/m)	Vapour pressure (N/m^2)
			Poise	kg/ms	Stoke	m^2/s			
Air	1.208	0.01185	1.85×10^{-4}	1.85×10^{-5}	1.53×10^{-1}	1.53×10^{-5}	–	–	–
Benzene	860	8.434	0.007	7.00×10^{-4}	8.14×10^{-3}	8.14×10^{-7}	1.0356×10^9	0.0255	1.000×10^4
Castor oil	960	9.414	9.800	9.80×10^{-1}	1.00×10^1	1.00×10^{-3}	1.441×10^9	0.0392	–
Carbon tetrachloride	1594	15.632	0.010	1.00×10^{-3}	6.04×10^{-3}	6.04×10^{-7}	1.104×10^9	0.0265	1.275×10^4
Ethyl alcohol	789	7.737	0.012	1.20×10^{-3}	1.52×10^{-2}	1.52×10^{-5}	1.118×10^9	0.0216	5.786×10^3
Glycerine	1260	12.356	8.350	8.35×10^{-1}	6.63	6.63×10^{-4}	4.354×10^9	0.0637	1.373×10^{-2}
Kerosene	800	7.845	0.020	2.00×10^{-5}	2.50×10^{-2}	2.50×10^{-5}	–	0.0235	–
Mercury	13550	132.880	0.016	1.60×10^{-3}	1.18×10^{-3}	1.18×10^{-7}	2.431×10^{10}	0.510	1.726×10^{-1}